GRE Quantitative Reasoning

모의고사 Set 3 — 문제지

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구성
Section 1: 12문항 / 21분
Section 2: 15문항 / 26분
합계 27문항 / 47분 · 계산기 사용 가능

대소비교(Quantitative Comparison) 선택지 — 모든 QC 문제 공통
(A) Quantity A is greater. (B) Quantity B is greater.
(C) The two quantities are equal. (D) The relationship cannot be determined from the information given.

안내
- Indicate all (복수선택) 문제는 해당하는 것을 모두 고르세요. 모두 맞아야 정답입니다.
- 주관식 분수 답은 약분하지 않아도 됩니다.
- 문제 이미지 속 번호는 문제집 원번호입니다. 문항 번호는 보라색 'Question N'을 기준으로 합니다.
- 정답과 해설은 별도 파일(Set 3 해설지)에 있습니다.

답안 기록표

S1	1	2	3	4	5	6	7	8	9	10	11	12
답												

S2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
답															

Section 1 12 Questions · 21 Minutes

Question 1 대소비교

500. Quantity A: The remainder when 11,999,999 is divided by 12
Quantity B: 11

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 2 대소비교

1230. In the xy-plane, S is the set of all points (c, d) for which c and d are both integers.
Quantity A: The number of points in S that are a distance of 2 units from the point (7, 5)
Quantity B: 4

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

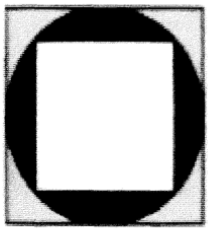
Question 3 대소비교

586. x and y are different positive integers
$x^2y=180$
Quantity A: x
Quantity B: y

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 4 대소비교

1009.



In the figure shown, a square is inscribed in a circle, which is inscribed in a square.
Quantity A: The total area of the regions shaded black
Quantity B: The total area of the regions shaded gray

(A) Quantity A is greater.

(B) Quantity B is greater.

(C) The two quantities are equal.

(D) The relationship cannot be determined from the information given.

Question 5 객관식

1173. If n is an integer greater than 2, which of the following could be a prime number?
A. $2n$
B. n^2
C. n^2+1
D. n^2-1
E. n^2+2n+1

(A) (B) (C) (D) (E)

Questions 6 to 8 are based on the following data.



Question 6 객관식

18. In 2003, what was the ratio of the number of in-store purchases paid for either by check or by credit card to the number of in-store purchases paid for in cash?

A. 52 to 33
B. 36 to 31
C. 33 to 31
D. 31 to 21
E. 12 to 11

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 7 객관식

19. If the distribution of the number of in-store purchases in 2003 by method of payment is to be represented by a circle graph with a radius of 3.2 centimeters, approximately what will be the length, in centimeters, of the arc of the sector representing payment by credit card?

A. 4.2
B. 4.8
C. 5.4
D. 6.0
E. 6.4

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 8 주관식

20. In 2003, what percent of the in-store purchases that were not paid for in cash were paid for by check?
Give your answer to the nearest whole percent.

[ ] %

Question 9 객관식

861~863

Percent of Country *M*'s Population by Region and Blood Type

Region of Country *M*	Blood Type: *O*-Rh⁻	*O*-Rh⁺	*A*-Rh⁻	*A*-Rh⁺	*B*-Rh⁻	*B*-Rh⁺	*AB*-Rh⁻	*AB*-Rh⁺	Total
R_1	4.8	0.8	5.7	1.0	1.1	0.2	1.3	0.1	15.0
R_2	15.4	1.2	14.2	1.4	1.6	0.8	1.1	0.1	35.8
R_3	14.1	2.5	15.3	2.4	4.0	0.7	0.7	0.2	39.9
R_4	1.0	0.6	2.6	0.8	0.8	0.5	0.3	0.1	6.7
R_5	0.8	0.3	0.9	0.2	0.1	0.1	0.2	*	2.6
Total	36.1	5.4	38.7	5.8	7.6	2.3	3.6	0.5	100.0

*less than 0.05

Total Population of Country *M*: 2,011,000

861. If a person is randomly chosen from Country M, approximately what is the probability that the person will have one of the two most common of the eight blood types?
A 7/8
B 4/5
C 3/4
D 5/8
E 3/5

A B C D E

Question 10 주관식

204. At a certain high school, 90 percent of seniors who take physics also take calculus and 60 percent of seniors who take calculus also take physics. If 30 percent of seniors at the school take physics, what percent of seniors take calculus?

[ ] %

Question 11 복수선택

382. Larry and Tony work for different companies. Larry`s salary is the 90th percentile of the salaries in his company, and Tony`s salary is the 70th percentile of the salaries in his company.
Which of the following statements individually provide(s) sufficient additional information to conclude that Larry`s salary is higher than Tony`s salary?
Indicate all such statements.
AThe average (arithmetic mean) salary in Larry`s company is higher than the average salary in Tony`s company
BThe median salary in Larry`s company is equal to the median salary in Tony`s company
CThe 80th percentile in Larry`s company is higher than the 70th percentile salary in Tony`s company

Indicate all that apply. A B C

Question 12 객관식

174. A candy shop owner decides to create a mixed assortment of candies by mixing two kinds of candies. One kind costs the owner 90 cents per pound, while another costs the owner 70 cents per pound. If the mixed assortment of candies costs the owner 85 cents per pound, and 30 pounds of 90 cents kind is used in the process, then what is the total pounds of mixed candies that can be created?
A40
B45
C50
D55
E60

A B C D E

Section 2 15 Questions · 26 Minutes

Question 1 대소비교

181. The retail price of a certain skirt was x percent greater than its wholesale price, where 20≤x≤80. The discount price of the skirt was x percent less than its retail price.
Quantity A: The wholesale price of the skirt
Quantity B: The discount price of the skirt

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 2 대소비교

109. Each of the people who took a data entry test made 0, 1, 2, 3, or 4 errors. Of those people, 65 percent made fewer than 4 errors, 46 percent made fewer than 3 errors, and 24 percent made fewer than 2 errors.
Quantity A: The percent of the people who took the data entry and made exactly 2 errors
Quantity B: 20%

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 3 대소비교

173. $x > y > z$
Quantity A: xy
Quantity B: yz

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 4 대소비교

928. The range of a list of 13 numbers is 110. The range of 10 of these numbers is 80.
Quantity A: The range of the remaining 3 numbers
Quantity B: 30

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 5 대소비교

1032. y > 1,001
Quantity A: $\sqrt[3]{y}$
Quantity B: y/100

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 6 객관식

1056. a_1, a_2, a_3, ..., a_n, ... The sequence of numbers shown is defined by $a_1=4$, $a_2=5$, and $an=2a_{n-1}+a_{n-2}$ for all integers greater than 2. Which of the following is a number in the sequence?
A9
B13
C18
D23
E33

(A) (B) (C) (D) (E)

Question 7 객관식

967. A student has a box containing only two kinds of bills: 6 Canadian dollar bills and y Australian dollar bills. If more than 3/8 of the bills in the box are Australian dollar bills, what is the least number of bills that the student could have in the box?
A7
B9
C10
D11
E12

(A) (B) (C) (D) (E)

Question 8 객관식

977. In a barrel, there are in ping-pong balls labeled with consecutive integers from 1 to n, where n is an odd integer greater than 50. If a ball is to be randomly selected from the barrel, what is the probability that the ball selected will be a ball that is labeled with an odd integer?
A1/2
B2/n
C(n−1)/n
D(n−1)/2n
E(n+1)/2n

(A) (B) (C) (D) (E)

Question 9 복수선택

114. A store is shipping 7 items to a single address. The items can be packed into 1, 2 or 3 different containers for shipping. If a container is used for shipping, it must be packed with at least 2 items. Which of the following statements about the different combinations of packing must be true?
Indicate all such statements.
AThere will be exactly 3 items in one container.
BThere will be at most 3 items in each container.
CThere will be an odd number of items in one container.

Indicate all that apply. A B C

Question 10 주관식

112. In quadrant I of the xy-plane, a triangle has vertices at (r, r), (4r, r), and (r, 3r). For what value of r is the area of the triangle equal to 300?

Questions 11 to 13 are based on the following data.

Measures Planned to be Used to Offset Rising Energy Costs: Survey Results for 1,600 Companies

Measure	Percent of Companies
Absorb increased costs	48%
Consolidate debt	15%
Cut research and development	10%
Increase prices	41%
Negotiate raw-material costs	23%
Reduce workforce	18%
Trim advertising and marketing	12%

Question 11 객관식

21. An analyst estimates that of the companies surveyed that plan to increase prices, 75 percent will actually increase prices. According to the analyst's estimate, what is the number of companies surveyed that plan to increase prices but will not actually increase prices?

A. 164
B. 262
C. 381
D. 547
E. 640

A B C D E

Question 12 객관식

22. For the seven measures shown, what is the median of the numbers of companies surveyed that plan to use the measures?

A. 656
B. 548
C. 402
D. 368
E. 288

A B C D E

Question 13 객관식

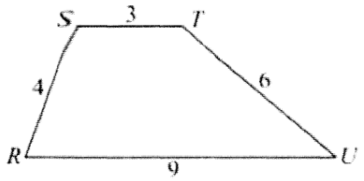
23. The number of companies surveyed that plan to trim advertising and marketing is what percent less than the number of companies surveyed that plan to consolidate debt?

A. 15%
B. 20%
C. 25%
D. 30%
E. 35%

A B C D E

Question 14 객관식

777.



The parallel sides (bases) of trapezoid RSTU have lengths 3 and 9, and the nonparallel sides have lengths 4 and 6, as shown in the figure. Line segment AB (not shown) is parallel to the two bases of RSTU with point A on side RS and point B on side TU. If AB divides the trapezoid into two trapezoids having equal perimeters, what is the sum of the lengths of the nonparallel sides of trapezoid RABU?
A2.0
B2.5
C3.0
D4.0
E5.0.

A B C D E

Question 15 복수선택

1082. The stock prices of three companies are x, 2x and x+2, respectively and x>2. Which of the following statement alone can be sufficient to determine the value of x?
Indicate all such statements.
AArithmetic Mean
BMedian
CRange

Indicate all that apply. A B C